UNIVERSITY OF THE PEOPLE

CS 2204-01 Communications and Networking -

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Assignment Activity Unit 2

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Question 1: CRC Error Detection and Correction

Introduction to CRC

Cyclic Redundancy Check (CRC) is one of the most powerful and widely adopted error-detection mechanisms used at the Data Link Layer of the OSI model. As a network technician in a data center, understanding how CRC works is fundamental to diagnosing frame transmission errors and maintaining data integrity across the network. According to Dordal (2023), the CRC error code is based on long division of polynomials where coefficients are integers modulo 2, and this mathematical foundation makes it extremely effective at catching burst errors that commonly occur in network transmission environments.

CRC operates by treating the message in binary as a polynomial, then performing a modulo-2 division using a predetermined generator polynomial. The remainder produced by this division is appended to the data frame before transmission. At the receiver side, the same division is

performed on the received frame. If the remainder equals zero, no error is detected; if the remainder is non-zero, an error has occurred during transmission.

Scenario Analysis

In this scenario, a data frame of **1,200 bits** was transmitted using the generator polynomial $\mathbf{x^4 + x^3 + x^2 + 1}$. After performing the CRC calculation at the receiver, a remainder of **1011** was obtained instead of the expected remainder of 0000.

Step 1: Interpreting the Generator Polynomial

The generator polynomial $x^4 + x^3 + x^2 + 1$ must first be converted to its binary representation. Each term corresponds to a bit position:

```
x⁴ --> position 4 --> 1
x³ --> position 3 --> 1
x² --> position 2 --> 1
x¹ --> position 1 --> 0 (no x¹ term)
x⁰ --> position 0 --> 1
```

This yields the binary divisor: **11101**. The degree of the polynomial is 4 (N = 4), which means 4 zero-bits are appended to the original data before performing the CRC division, and the resulting CRC remainder will be 4 bits in length.

Step 2: Error Detection

At the receiver side, the incoming frame (original data bits + the 4 appended CRC bits) is divided by the same generator polynomial 11101 using modulo-2 arithmetic (XOR operations). If the data was transmitted without corruption, this division should yield a remainder of exactly 0000.

However, in our scenario, the receiver computed a remainder of **$1011 \neq 0000$**. Since the remainder is non-zero, this conclusively indicates that **the data frame contains a transmission error**. This is consistent with the CRC error detection principle described by Dordal (2023), which states that a corrupted packet yields a non-zero remainder upon division by p(X).

The non-zero remainder 1011 serves as the error syndrome. It confirms that one or more bits were corrupted during transmission across the data center network a situation that can arise from electrical interference, hardware faults, or medium noise.

Step 3: Error Correction Process

It is important to note that CRC by itself is an error-detection mechanism, not an error-correction mechanism. CRC confirms that an error exists but does not directly identify which bit(s) are

corrupted. This is a key distinction between CRC and error-correcting codes such as Hamming codes, which are designed to both detect and correct single-bit errors (Dordal, 2023).

The standard error correction procedure in a CRC-based system relies on the retransmission protocol. When the receiver detects a non-zero remainder, it discards the corrupted frame and signals the sender to retransmit. In this data center scenario, the following corrective steps are taken:

1. **Error** Identified: The remainder 1011 signals frame corruption.
2. Frame Discarded: The receiver drops the corrupted 1,200-bit frame.
3. **Retransmission** Triggered: A Negative Acknowledgment (NAK) or timeout triggers the sender to retransmit the frame.
4. **CRC** Re-appended: The sender recalculates the CRC for the data, appending a fresh 4-bit remainder.
5. **Correct** Frame Accepted: Upon successful retransmission, the receiver verifies remainder = 0000 and accepts the frame.

For example, if the original data is represented as D(x), the sender computes the CRC by dividing $x^4 \times D(x)$ by the generator polynomial G(x) = 11101. The remainder r(x) of maximum degree 3 (4 bits) is then appended to D(x) to form the transmitted frame $T(x) = x^4 \times D(x) + r(x)$. At the receiver, dividing T(x) by G(x) should yield remainder 0 for an error-free frame. Since our received remainder is 1011, the transmitted frame is corrupted and must be retransmitted.

CRC Limitations and Practical Considerations

While CRC is highly effective, it does have limitations. Dordal (2023) notes that CRC is not secure against intentional corruption; given an arbitrary message, there are algebraic means for tweaking the last bytes so that the CRC code equals any predetermined value. However, for unintentional transmission errors in data centers, CRC with a degree-4 polynomial provides $2^4 = 16$ possible remainders, offering strong detection capabilities for single-bit, double-bit, and burst errors up to 4 bits in length. For production data centers, Ethernet employs a 32-bit CRC (CRC-32), which provides even stronger protection with 2^{32} possible checksum values.

In summary, the data frame of 1,200 bits does contain a transmission error, as confirmed by the non-zero CRC remainder of 1011. The correct frame is recovered through retransmission, after which the sender recomputes the CRC, appends the correct 4-bit FCS (Frame Check Sequence) to

the data, and the receiver verifies a zero remainder upon receipt. This demonstrates the critical role CRC plays in maintaining data integrity at the Data Link Layer of the OSI model.

Question 2: Selective Repeat ARQ

Introduction to Selective Repeat ARQ

Automatic Repeat Request (ARQ) protocols are essential error-control mechanisms at the Data Link Layer that ensure reliable data delivery over unreliable communication channels. Among the ARQ variants, Selective Repeat ARQ (SR-ARQ) is the most efficient because it requires retransmission of only the specific frames that were lost or corrupted, rather than retransmitting all subsequent frames as the Go-Back-N ARQ does (Simplilearn, 2022).

In the SR-ARQ protocol, both the sender and receiver maintain a window of frames. The sender can transmit multiple frames within the window size without waiting for individual acknowledgments. The receiver, however, individually acknowledges each successfully received frame and buffers out-of-order frames while waiting for missing ones. This selective approach maximizes channel utilization and reduces unnecessary retransmissions.

Scenario Setup

In this scenario, the sender transmits a window of 8 frames numbered from 0 to 7 using Selective Repeat ARQ. The transmission status is as follows:

```
Window size (N) = 8
Frame sequence numbers: 0, 1, 2, 3, 4, 5, 6, 7
Frames successfully received: 2, 4, 7
Frames NOT received (lost/corrupted): 0, 1, 3, 5, 6
```

It is important to note that for Selective Repeat ARQ, the maximum window size must satisfy $N \leq 2^{(k-1)}$, where k is the number of bits in the sequence number. With 8 frames (sequence numbers 0-7), we need $k = 3$ bits, so $N \leq 4$. However, for the purposes of this problem analysis, we will work with the given window of 8 frames and the stated reception pattern.

Step 1: Calculating the Cumulative Acknowledgment Number

In Selective Repeat ARQ, the receiver sends individual acknowledgments (ACKs) for each frame it receives correctly, regardless of order. This is a key distinction from Go-Back-N ARQ, which uses cumulative acknowledgments. The receiver buffers out-of-order frames and waits for missing frames to arrive.

The cumulative acknowledgment number represents the sequence number up to which all frames have been received in order (i.e., no gaps). This is computed by examining the sequence of received frames starting from frame 0:

```
Frame 0: NOT received <-- first gap detected here
Frame 1: NOT received
Frame 2: Received successfully
Frame 3: NOT received
Frame 4: Received successfully
Frame 5: NOT received
Frame 6: NOT received
Frame 7: Received successfully
```

Starting from frame 0, since frame 0 was NOT received, there is an immediate gap at the beginning of the sequence. The cumulative acknowledgment number is the sequence number of the last frame received in an unbroken sequence starting from the lowest expected frame. Since frame 0 was not received, **no frames have been received in a continuous sequence from the start**. Therefore, the cumulative ACK number sent back to the sender is **ACK 0** (indicating the receiver is still expecting frame 0, the next frame it needs).

In SR-ARQ, while the cumulative ACK remains at 0 (because frame 0 is missing), the receiver also sends individual positive acknowledgments (ACKs) for the out-of-order frames it has successfully buffered. Therefore, the receiver sends:

```
Individual ACK for Frame 2 (buffered, waiting for frames 0 and 1)
Individual ACK for Frame 4 (buffered, waiting for frame 3)
Individual ACK for Frame 7 (buffered, waiting for frames 5 and 6)
Cumulative ACK = 0    (requesting retransmission starting from
frame 0)
```

Step 2: Calculating the Expected Next Sequence Number

The expected next sequence number for transmission refers to the next frame the sender needs to retransmit or send. In Selective Repeat ARQ, the sender must retransmit only the frames that were not acknowledged. Based on the scenario:

Frames that were NOT successfully received (requiring retransmission): 0, 1, 3, 5, and 6. The sender's window does not slide forward until frame 0 (the base of the window) is acknowledged. Since the receiver has signaled that it is waiting for frame 0 through the cumulative ACK = 0, the sender must first retransmit frame 0.

Therefore, the **expected next sequence number for transmission = 0**, as the sender must retransmit frame 0 to allow the window to advance. Once frame 0 is successfully delivered and

acknowledged, the cumulative ACK advances, and the sender then retransmits frame 1, followed by frame 3, then frames 5 and 6, one by one — as required by the Selective Repeat mechanism.

Efficiency Analysis: SR-ARQ vs. Go-Back-N ARQ

The efficiency advantage of Selective Repeat ARQ over Go-Back-N ARQ is clearly demonstrated in this scenario. Undergo-Back-N ARQ, when frame 0 fails, the sender would be forced to retransmit ALL 8 frames (0 through 7), discarding the successfully received frames 2, 4, and 7. This results in significant wasted bandwidth.

Under Selective Repeat ARQ, only the 5 missing frames (0, 1, 3, 5, 6) need to be retransmitted, while frames 2, 4, and 7 are buffered at the receiver and do not need to be resent. As Simplilearn (2022) explains, SR-ARQ only retransmits the damaged or lost frame, which significantly reduces the number of retransmissions and increases the overall efficiency of the network model. The bandwidth requirement is also moderate in SR-ARQ compared to the excessively high bandwidth demands of Go-Back-N when many retransmissions are required.

However, SR-ARQ does require a larger buffer at both the sender and receiver sides to store multiple frames simultaneously. This is a trade-off: higher memory usage in exchange for greater channel efficiency and reduced retransmission overhead — a trade-off that is generally favorable in high-speed data center environments where bandwidth is valuable and memory is relatively inexpensive.

Summary of Results

```
Window size: 8 frames (sequence numbers 0 to 7)
Successfully received frames: 2, 4, 7
Lost/corrupted frames requiring retransmission: 0, 1, 3, 5, 6
Cumulative ACK number sent to sender: ACK 0
Expected next sequence number for transmission: 0
Individual ACKs sent for buffered frames: ACK 2, ACK 4, ACK 7
```

In conclusion, the Selective Repeat ARQ mechanism in this data center scenario demonstrates its superior efficiency by isolating only the erroneous frames for retransmission. The cumulative acknowledgment number is 0, indicating that the receiver is waiting for frame 0 as the first in-sequence frame, and the expected next sequence number for retransmission is also 0. This targeted approach to error recovery ensures reliable data delivery while minimizing channel waste, making SR-ARQ the preferred error-control protocol in bandwidth-sensitive networking environments.

References

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